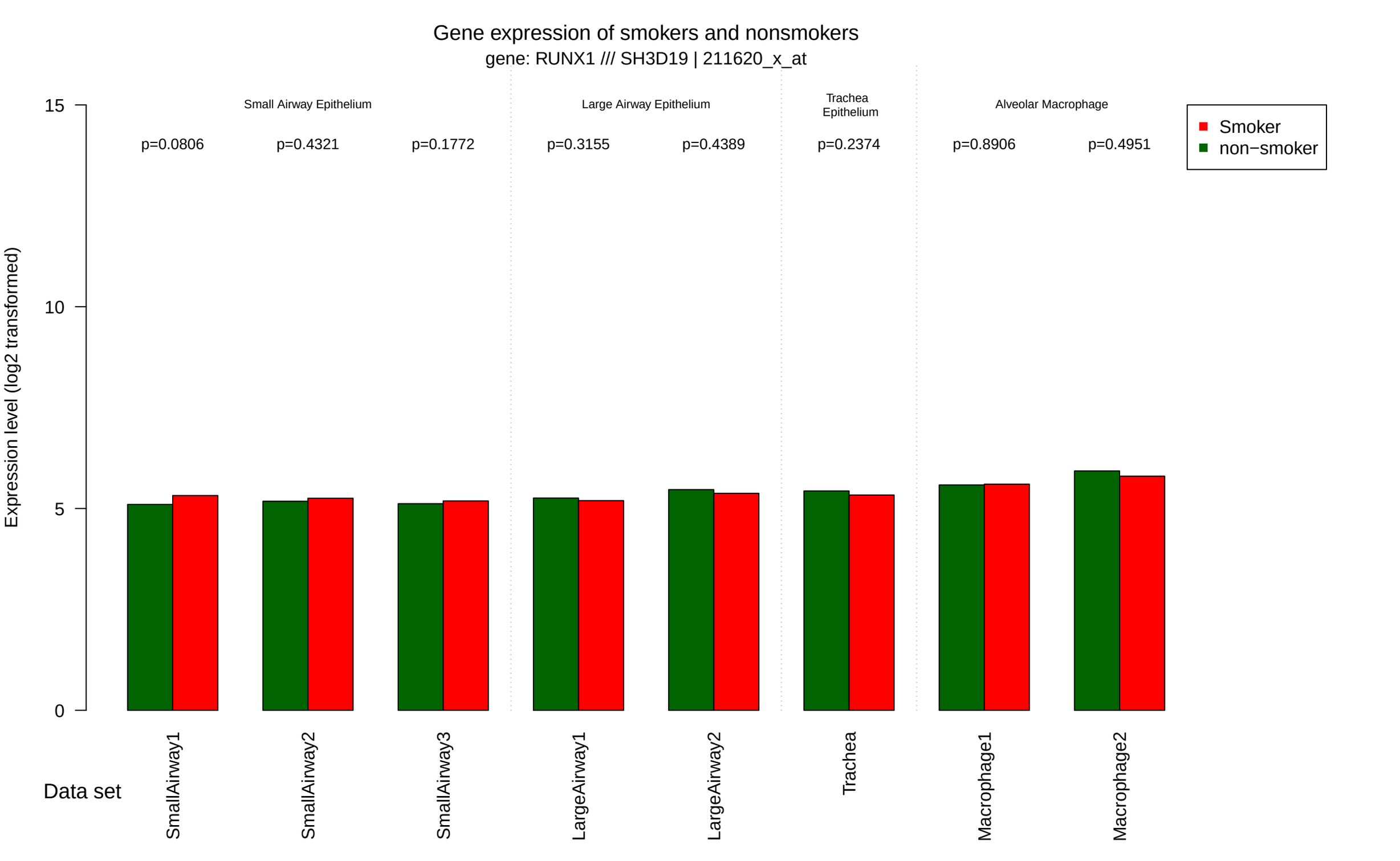
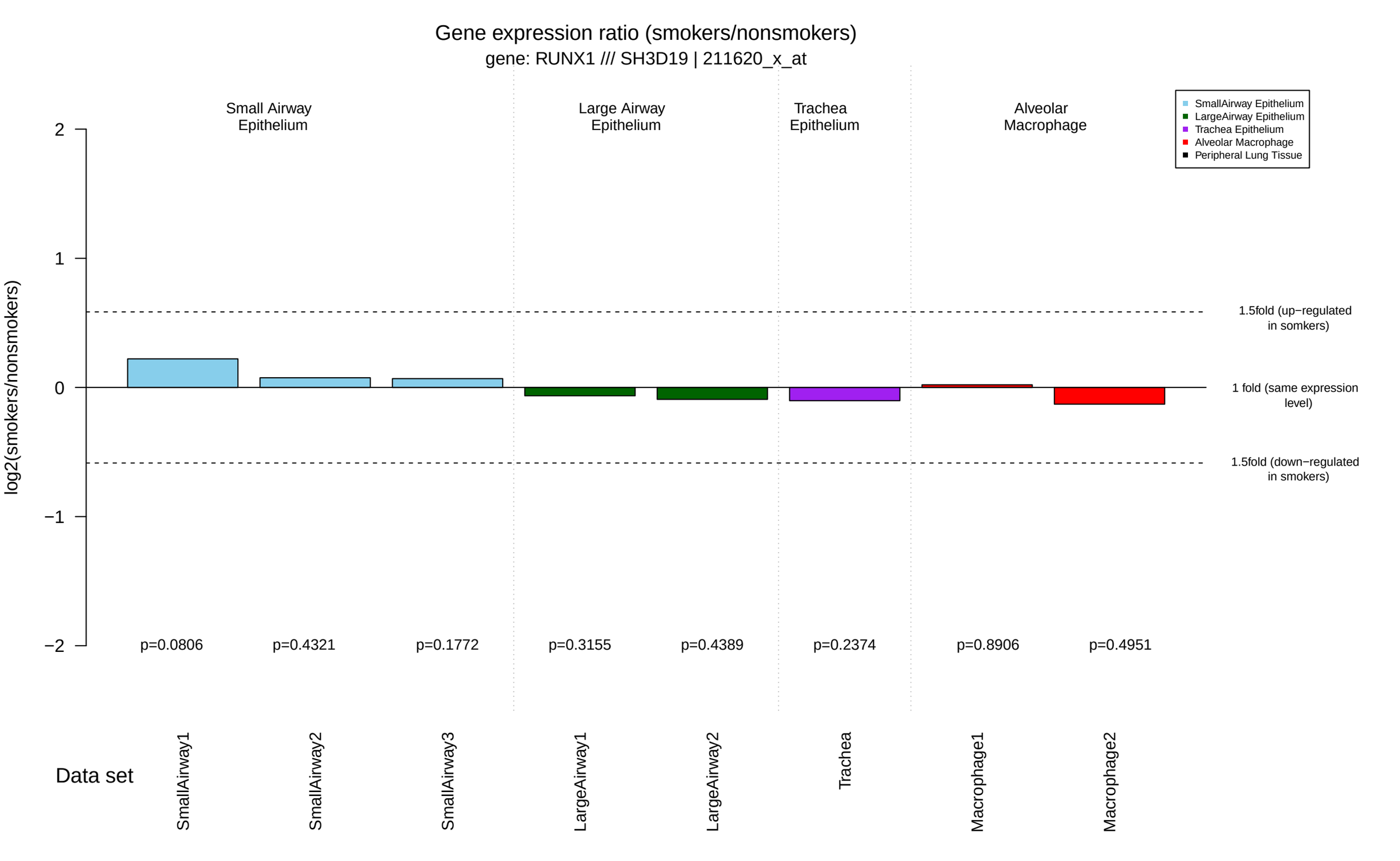
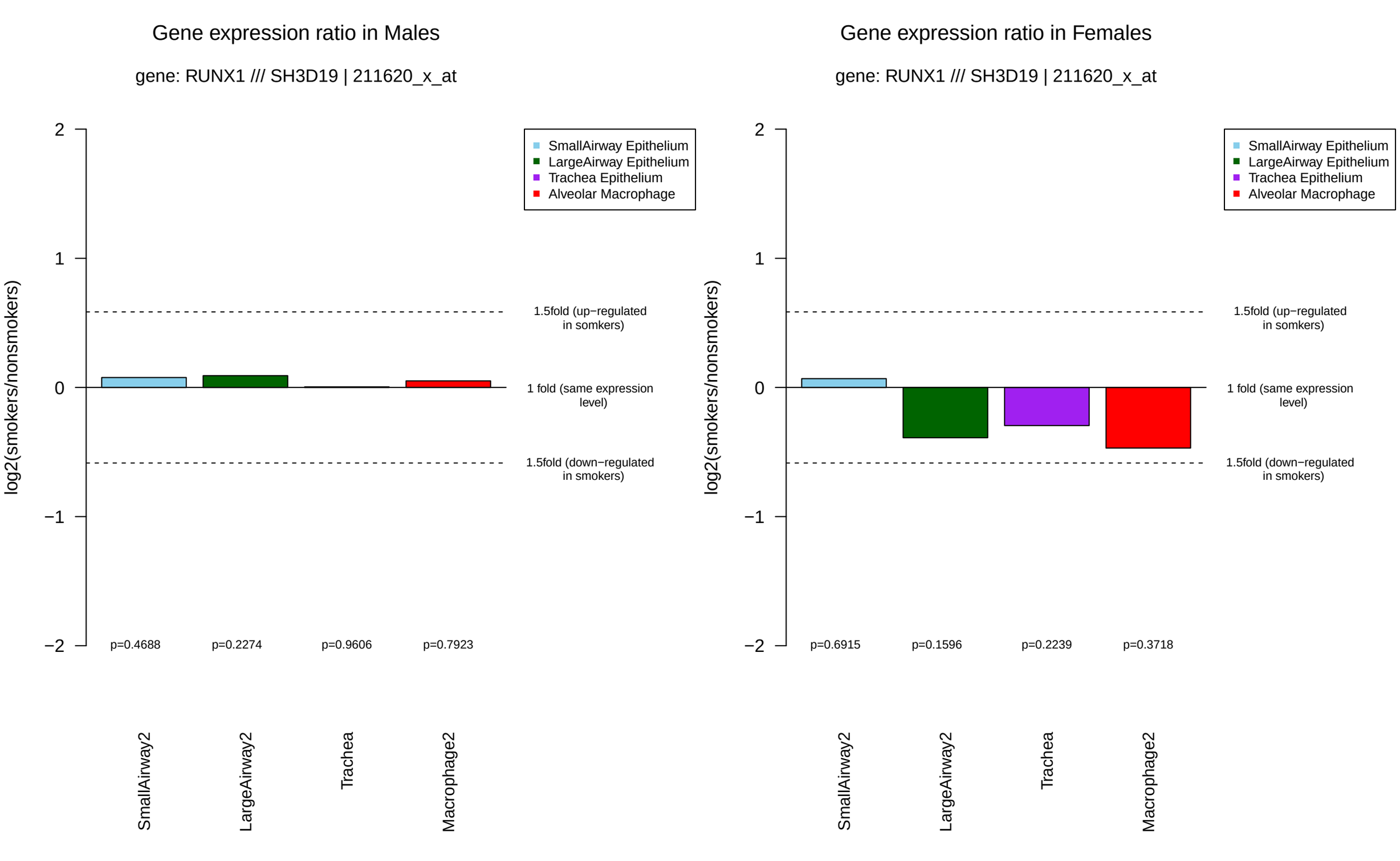
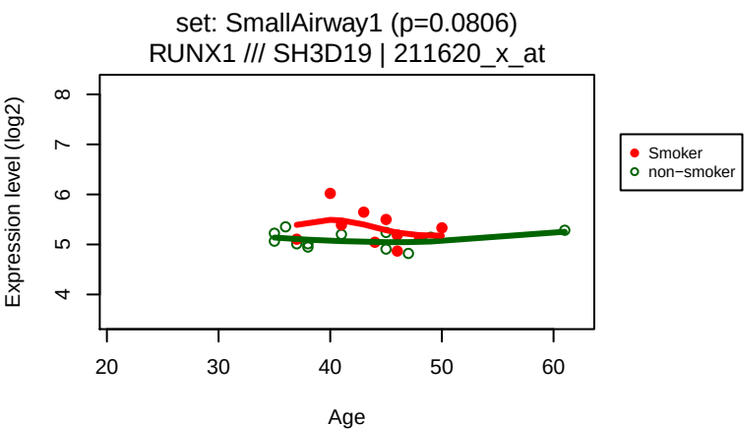
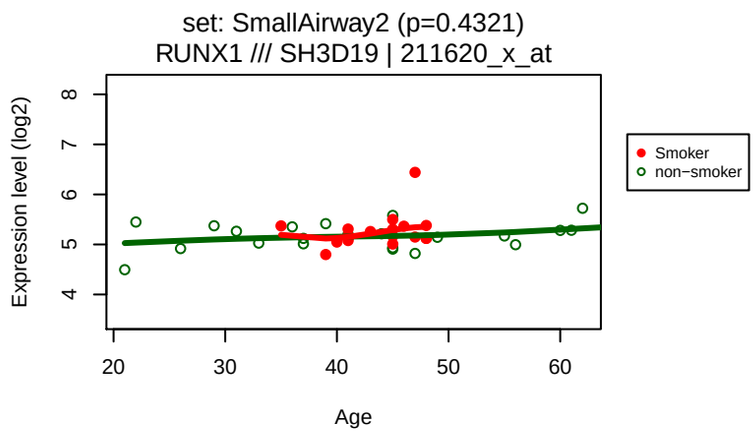
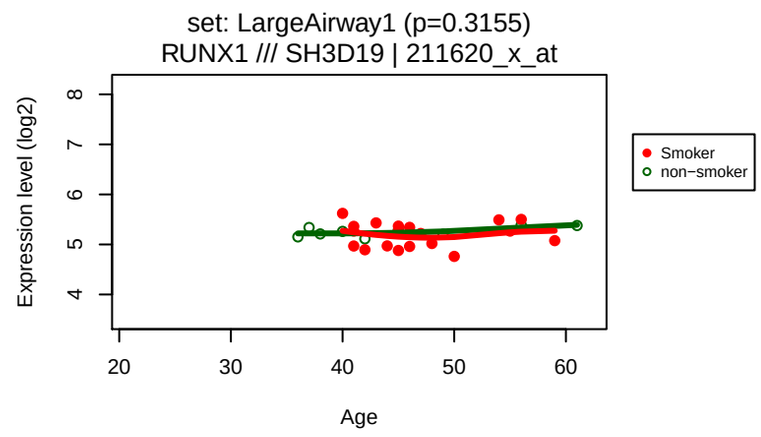

Gene expression of smokers and nonsmokers
gene: RUNX1 /// SH3D19 | 211620_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0806
p=0.4321
p=0.1772
p=0.3155
p=0.4389
p=0.2374
p=0.8906
p=0.4951
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

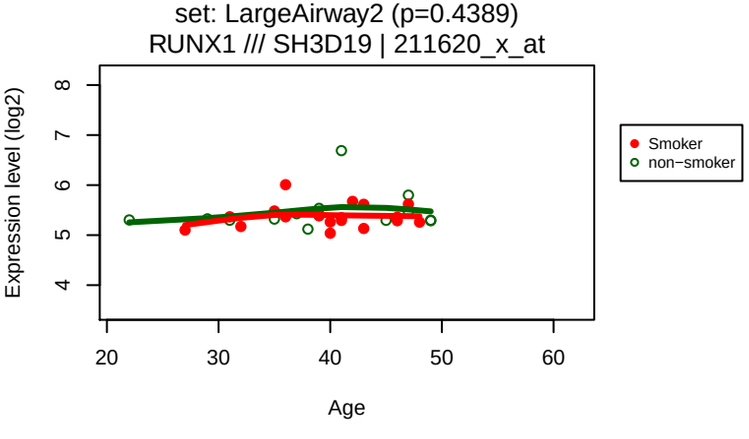

Gene expression ratio (smokers/nonsmokers)
gene: RUNX1 /// SH3D19 | 211620_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0806
p=0.4321
p=0.1772
p=0.3155
p=0.4389
p=0.2374
p=0.8906
p=0.4951
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

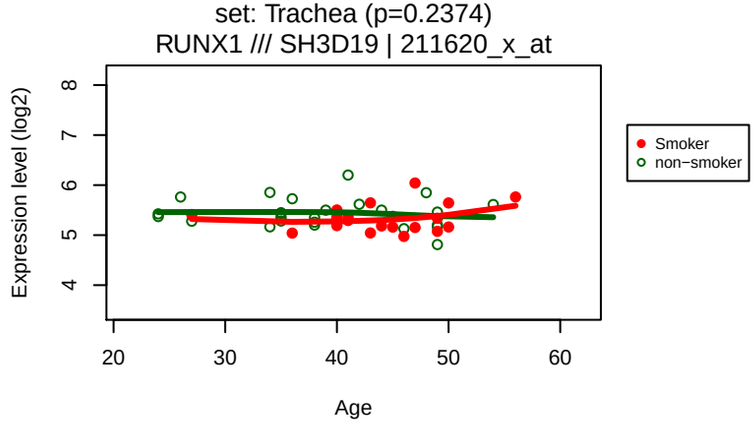

Gene expression ratio in Males
gene: RUNX1 /// SH3D19 | 211620_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4688
p=0.2274
p=0.9606
p=0.7923
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RUNX1 /// SH3D19 | 211620_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6915
p=0.1596
p=0.2239
p=0.3718
SmallAirway2
LargeAirway2
Trachea
Macrophage2

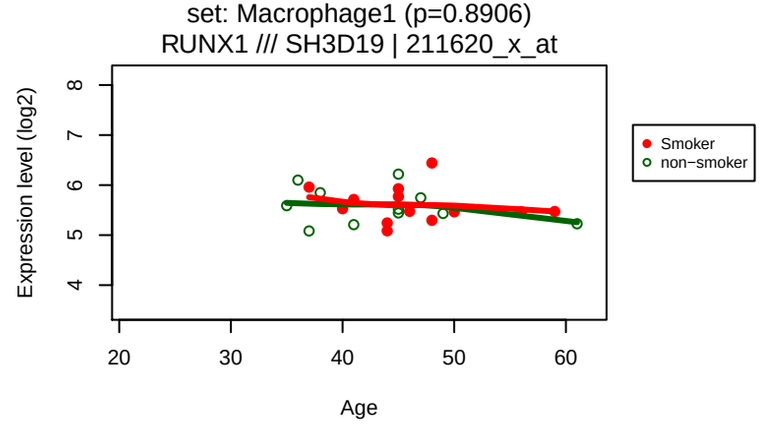

set: SmallAirway1 (p=0.0806)
RUNX1 /// SH3D19 | 211620_x_at
Expression level (log2)
Age
Smoker
non−smoker

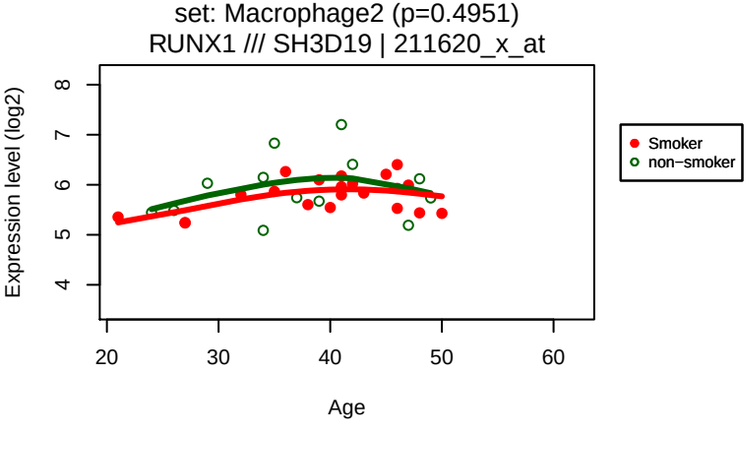

set: SmallAirway2 (p=0.4321)
RUNX1 /// SH3D19 | 211620_x_at
Expression level (log2)
Age
Smoker
non−smoker

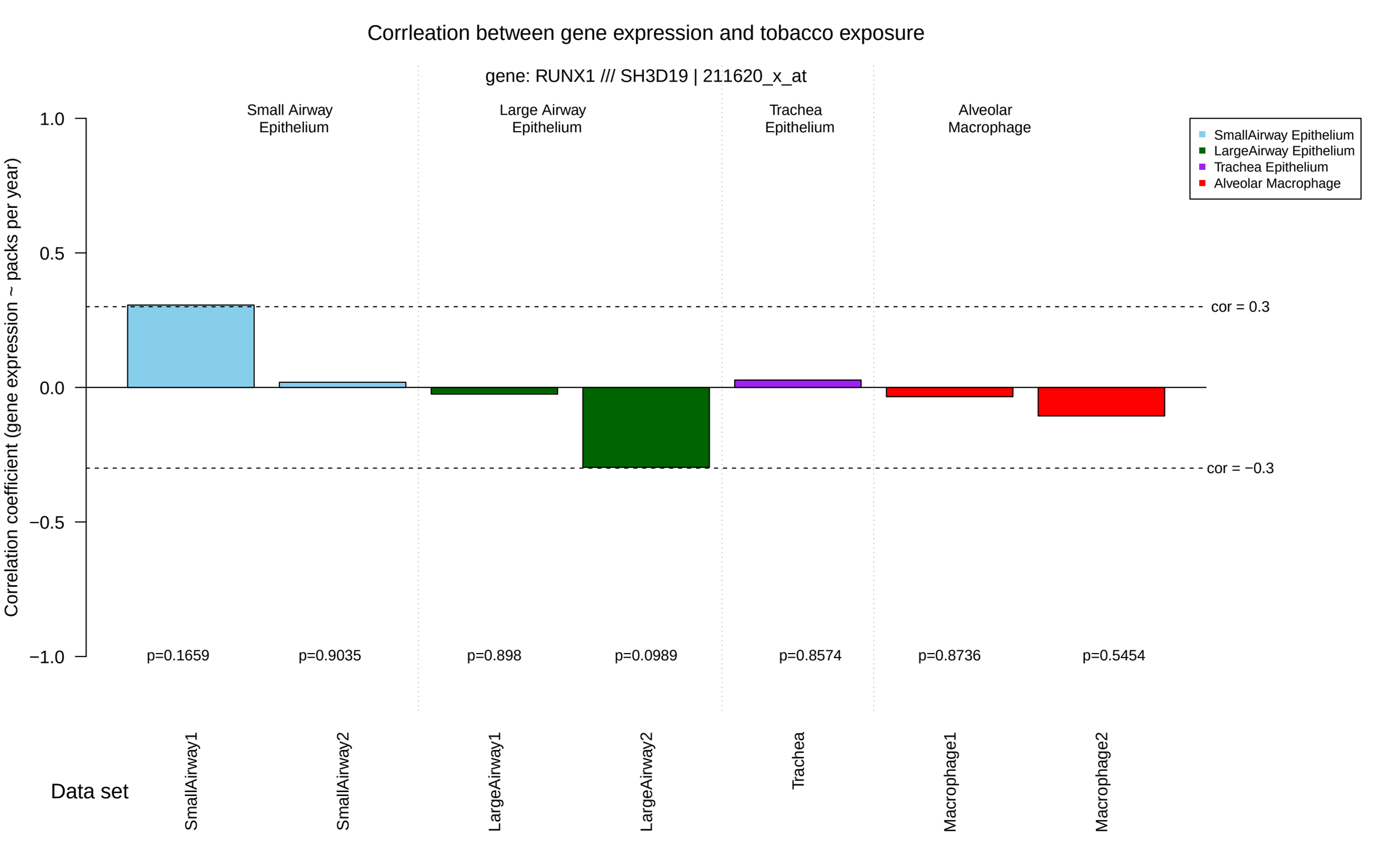

set: LargeAirway1 (p=0.3155)
RUNX1 /// SH3D19 | 211620_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4389)
RUNX1 /// SH3D19 | 211620_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2374)
RUNX1 /// SH3D19 | 211620_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8906)
RUNX1 /// SH3D19 | 211620_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4951)
RUNX1 /// SH3D19 | 211620_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RUNX1 /// SH3D19 | 211620_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1659
p=0.9035
p=0.898
p=0.0989
p=0.8574
p=0.8736
p=0.5454
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2